

Goal. Explore even more methods of determining whether series converge or diverge.

1. Name all the tests for convergence/divergence of series in our series toolbox. Put a * by the tests that only work for series with positive terms.
Note, your answer should be longer than last worksheet!

2. Use any of the tests you listed in 1. to determine whether the following series converge or diverge. Be sure to verify the hypotheses, state your conclusion, and say which test you used. If you can, list a second test that might work as well.

(a)
$$\sum_{n=1}^{\infty} \frac{(-1)^n}{4^{2n+1}(n+1)}$$

(b)
$$\sum_{n=1}^{\infty} \frac{e^n}{n!}$$

(c)
$$\sum_{n=3}^{\infty} \frac{(-12)^n}{n}$$

(d) $\sum_{n=1}^{\infty} \frac{5^n}{n^n}$

(Hint: When you see something to the n th power in a series, it's often a good idea to use the root test.)

(e) $\sum_{n=1}^{\infty} \left(2^{-n} + \frac{3n+5}{n^2} \right)$

(f) $\sum_{n=2}^{\infty} \frac{n}{(n^3 - 1)^{\frac{3}{7}}}$